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COMPUTER SCIENCE

9618/23

Paper 2 Fundamental Problem-solving and Programming Skills

May/June 2023

2 hours

You must answer on the question paper.

You will need: Insert (enclosed)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- You may use an HB pencil for any diagrams, graphs or rough working.
- Calculators must **not** be used in this paper.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].
- No marks will be awarded for using brand names of software packages or hardware.
- The insert contains all the resources referred to in the questions.

This document has **20** pages. Any blank pages are indicated.

Refer to the **insert** for the list of pseudocode functions and operators.

- 1 The following pseudocode represents part of the algorithm for a program.

Line numbers are for reference only.

```

10  DECLARE Sheet4 : ARRAY[1:2, 1:50] OF INTEGER
...
100 FOR PCount ← 0 TO 49
101     Sheet4[1, PCount] ← 0
102     Sheet4[2, PCount] ← 47
103 NEXT PCount

```

- (a) The pseudocode contains references to an array.

Complete the table by writing the answer for each row.

Answer

The dimension of the array	
The name of the variable used as an array index	
The number of elements in the array	

[3]

- (b) The pseudocode contains two errors. One error is that variable `PCount` has not been declared.

Identify the **other** error **and** state the line number where it occurs.

Error

.....

.....

Line number

[2]

- (c) The pseudocode does not include a declaration for `PCount`.

State the data type that should be used in the declaration.

..... [1]

- (d) The pseudocode statements given in the following table are used in other parts of the algorithm.

Complete the table by placing **one or more** ticks (✓) in each row.

The first row has already been completed.

Pseudocode statement	Input	Process	Output
INPUT MyChoice	✓		
OUTPUT FirstName & LastName			
WRITEFILE YourFile, TextLine			
READFILE MyFile, TextLine			
Result $\leftarrow$ SQRT (NextNum)			

[4]

2 A program stores a date of birth for a student using a variable, `MyDOB`, of type `DATE`.

(a) `MyDOB` has been assigned a valid value corresponding to Kevin's date of birth.

Complete the pseudocode statement to test whether Kevin was born on a Thursday.

IF THEN [2]

(b) A function `CheckDate()` will take three integer parameters representing a day, month and year of a given date.

The function will validate the date of birth for a student that the parameters passed to it represent.

For a date to be valid, a student must be at least 18 in year 2020.

(i) Two of the parameter values can be checked without reference to the third parameter.

Describe these **two** checks.

Check 1

Check 2

 [2]

(ii) Several values of the parameter representing the day can only be checked completely by referring to the value of **one other** parameter.

Describe this check.

.....

 [2]

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3 A program processes data using a stack. The data is copied to a text file before the program ends.

(a) The following diagram shows the current state of the stack.

The operation of this stack may be summarised as follows:

- The `TopOfStack` pointer points to the last item added to the stack.
- The `BottomOfStack` pointer points to the first item on the stack.
- The stack grows upwards when items are added.

Stack		Pointer
Memory location	Value	
506		
505	WWW	← <code>TopOfStack</code>
504	YYY	
503	XXX	
502	ZZZ	
501	NNN	
500	PPP	← <code>BottomOfStack</code>

(i) An error will be generated if an attempt is made to POP a value when the stack is empty.

State the maximum number of consecutive POP operations that could be performed on the stack shown above **before** an error is generated.

..... [1]

(ii) The following operations are performed:

1. POP and store value in variable `Data1`
2. POP and store value in variable `Data2`
3. PUSH value AAA
4. PUSH value BBB
5. POP and discard value
6. POP and store value in variable `Data2`

Complete the diagram to show the state of the stack and the variables **after** the given operations have been performed.

Stack		Pointer	
Memory location	Value		
506			
505			
504			
503			
502			
501			
500			

Variable	Value
Data1	
Data2	

[4]

(b) The data is copied to a text file before the program ends.

- (i) State an advantage of writing the data from the stack to a text file before the program ends.

.....
 [1]

- (ii) A module `SaveStack()` will write the data from the stack to a text file.

Express an algorithm for `SaveStack()` as five steps that could be used to produce pseudocode.

Write the **five** steps.

Step 1

.....

Step 2

.....

Step 3

.....

Step 4

.....

Step 5

.....

[5]

4 A function `MakeString()` will:

1. take two parameters:
 - a count as an integer
 - a character
2. generate a string of length equal to the count, made up of the character
3. return the string generated, or return "ERROR" if the count is less than 1.

For example, the function call:

`MakeString(3, 'z')` will return the string "zzz"

Write pseudocode for function `MakeString()`.

[6]

5 A program is designed, coded and compiled without errors. The compiled code is sent for testing.

(a) The program will be tested using the walkthrough method.

Additional information will be needed before this method can be used.

Identify this additional information **and** explain why it is needed.

Additional information

.....

Explanation

.....

.....

.....

[3]

(b) Testing is completed and the program is made available to users.

Some time later, changes are made to the program to improve the speed of response.

State the type of maintenance that has been applied to the program.

..... [1]

6 A procedure `Select()` will:

1. take two integer values as parameters representing start and end values where both values are greater than 9 and the end value is greater than the start value
2. output **each** integer value between the start and the end value (**not** including the start and end values), where the sum of the last two digits is 6, for example, 142.

(a) Write pseudocode for procedure `Select()`.

Parameter validation is **not** required.

[7]

- (b) The check performed by procedure `Select()` on the last two digits is needed at several places in the program and will be implemented using a new function.

The new function `CheckNum()` will:

- allow the required sum to be specified (not just 6)
- check one number
- return an appropriate value.

Describe the function interface **and two** advantages of this modular approach.

Interface

.....

.....

.....

.....

Advantage 1

.....

.....

Advantage 2

.....

[4]

- 7 A school has a library system which allows students to borrow books for a length of time. Information relating to students and books is stored in text files. Student information includes name, home address, email address, date of birth, tutor and subject choices. Book information includes author, title, subject category, library location and the date that the book was borrowed.

A program helps the staff to manage the borrowing of books.

- (a) A new module needs to be written to generate emails to send to students who have an overdue book. Students who are sent an email are prevented from borrowing any more books until the overdue book is returned.

The process of abstraction has been used when designing the new module.

- (i) State the purpose of applying abstraction to this problem.

.....
 [1]

- (ii) Identify **one** item of information that is required and **one** item that is **not** required in the new module. Justify your choices.

Item required

Justification

.....

Item not required

Justification

.....
 [2]

- (iii) Identify **two** operations that would be required to process data when an overdue book is returned.

Operation 1

.....

Operation 2

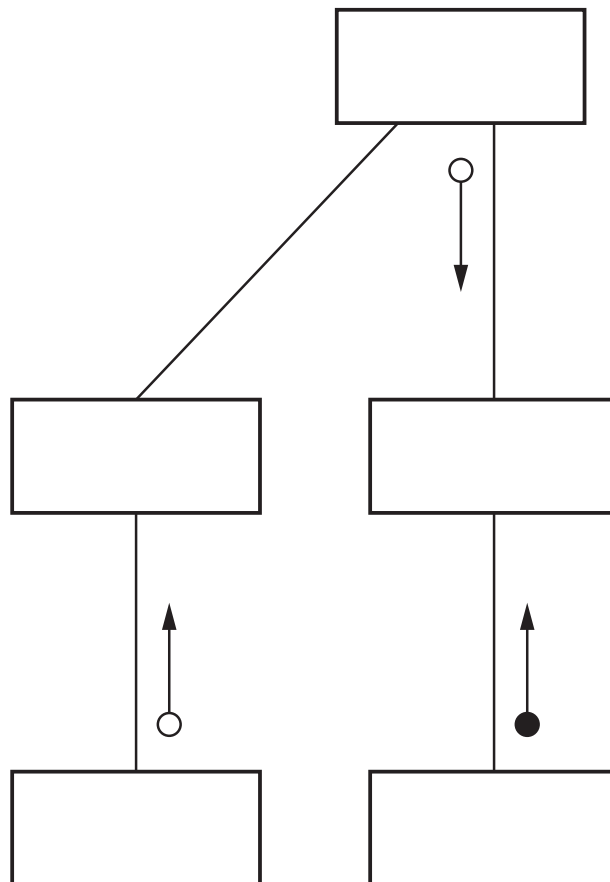
..... [2]

(b) Part of the library program contains program modules with headers as follows:

Pseudocode module header
PROCEDURE Module-X()
PROCEDURE Module-Y(BYREF RA : INTEGER, SA : REAL)
PROCEDURE Overlay()
FUNCTION Replace(RA : INTEGER, RB : BOOLEAN) RETURNS BOOLEAN
FUNCTION Reset(TA : STRING) RETURNS INTEGER

Module-X() and Module-Y() are both called from module Overlay().

Complete the structure chart.



[3]

- 8 A computer shop assembles desktop computers, using items bought from several suppliers. A text file `Stock.txt` contains information about each item.

Information for each item is stored as a single line in the `Stock.txt` file in the format:

`<ItemNum><SupplierCode><Description>`

Item information is as follows:

	Format	Comment
ItemNum	4 numeric characters	unique number for each item in the range "0001" to "5999" inclusive
SupplierCode	3 alphabetic characters	code to identify the supplier of the item
Description	a string	a minimum of 12 characters

The file is organised in ascending order of `ItemNum` and does not contain all possible values in the range.

The programmer has defined the first program module as follows:

Module	Description
<code>ChangeSupp()</code>	- called with two parameters <code>Code1</code> and <code>Code2</code> of type string that represent valid supplier codes - creates a new file <code>NewStock.txt</code> from the contents of the file <code>Stock.txt</code> where any reference to <code>Code1</code> is replaced by <code>Code2</code> - returns a count of the number of items that have had their supplier code changed

(a) Write pseudocode for module `ChangeSupp()`.

[8]

(b) A new module is required:

Module	Description
Report_1()	• takes a parameter of type string that represents a SupplierCode • searches the Stock.txt file for each line of item information that contains the given SupplierCode • produces a formatted report of items for the given SupplierCode, for example, for supplier DRG, the output could be: <pre> Report for Supplier: DRG Item Description 1234 USB Printer Cable 3m 1273 32GB USB Flash Drive 1350 Mouse Mat 320 x 240mm Number of items listed: 3 </pre>

Write pseudocode for module `Report_1()`.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins or other markings on the paper.

- (c) The format of the output from module `Report_1()` from **part (b)** is changed. The number of items listed is moved to the top of the report as shown in the example:

```
Report for Supplier: DRG
Number of items listed: 3
```

Item	Description
1234	USB Printer Cable 3m
1273	32GB USB Flash Drive
1350	Mouse Mat 320 x 240mm

- (i) Explain why this new layout would increase the complexity of the algorithm.

.....

.....

.....

..... [2]

- (ii) The algorithm will be modified to produce the report in the new format. The modified algorithm will be implemented so that the file `Stock.txt` is only read once.

Describe the modified algorithm.

.....

.....

.....

.....

.....

.....

..... [3]

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